

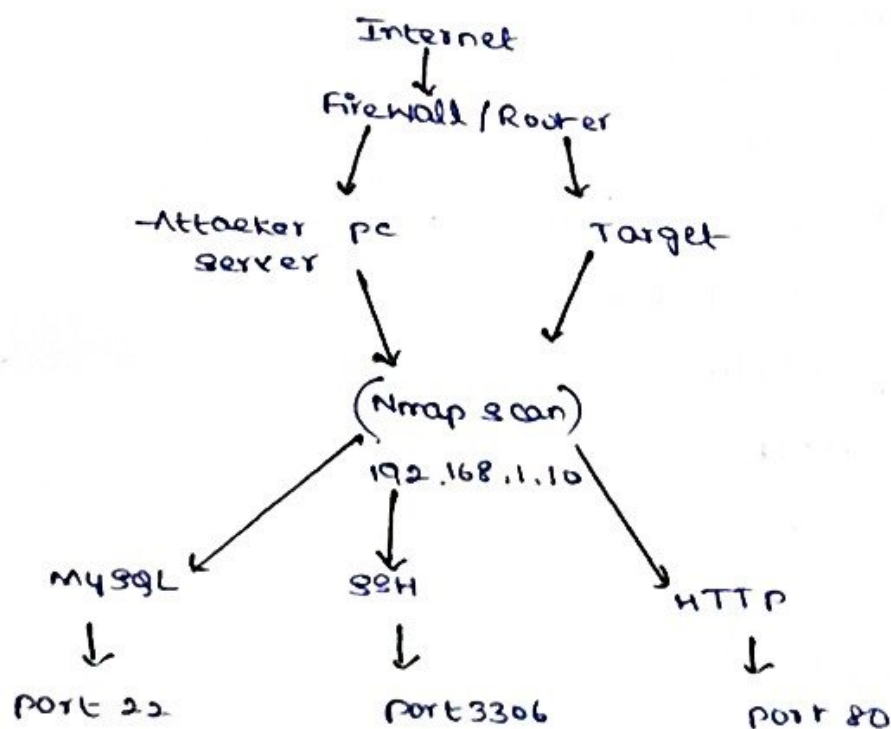
1) Network Scan Report Analysis using Nmap.

Introduction:

Network scanning is an important activity in ethical hacking used to identify live hosts, open ports, operating systems, and running services. Nmap (Network Mapper) is one of the most popular security assessment tools. In this report the target system 192.168.1.10 was scanned and the following services were identified

<u>Port</u>	<u>Service</u>	<u>Status</u>
22	SSH	open
80	HTTP	open
3306	MySQL	open

Network Diagram



1) Analysis of open ports

(A) Port 22 - SSH (Secure Shell)

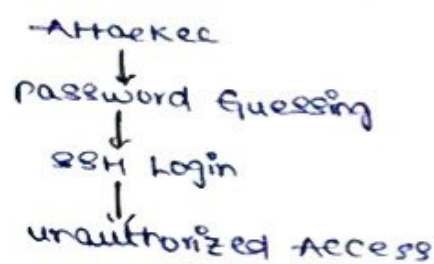
Purpose:

SSH is used for secure remote login and remote administration.

Security Risks

- Weak passwords
- Brute-force login attacks
- default usernames
- outdated SSH versions

Possible Attack



Protection

- Strong passwords
- Disable root login
- use SSH keys
- Enable Multi-Factor Authentication (MFA)
- change default port
- Install Fail2Ban

(B) Port 80 - HTTP

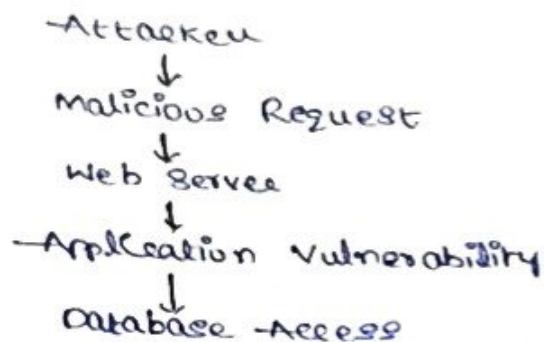
Purpose

provides web services through a browser

Security Risks

- SQL Injection
- Cross-Site Scripting (XSS)
- Directory Traversal
- File Upload Attacks

Attack Diagram



Protection:

- Use HTTPS
- Web Application Firewall
- Input validation
- Secure Coding
- Regular updates
- Disable directory listing.

C) Port 3306 - MySQL Database

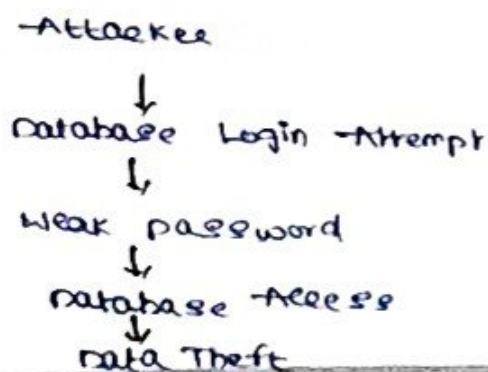
Purpose:

Stores application data

Security Risks

- Weak database password
- default mysql accounts
- Remote database access
- Data theft

Attack Diagram



Protection:

- Strong passwords
- Disable remote access
- Restrict IP address
- Database encryption
- Regular backups

Operating System Detection

Nmap can identify

- operating system
- Service versions
- Device type

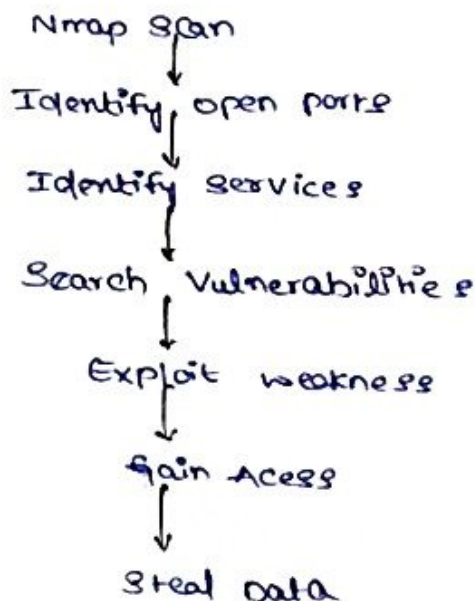
Risks

- Attackers identify vulnerabilities
- easier exploitation using known CVEs

Protection

- keeps OS updated
- Disable unnecessary services
- Apply security patches
- Enable firewall

Overall Attack Flow



Conclusion:

The Nmap scan identified three important services running on the target machine: SSH (22), HTTP (80), and MySQL (3306). If these services are not properly secured, attackers may exploit weak passwords, outdated software, or application vulnerabilities to gain unauthorized access or steal sensitive data. Implementing strong authentication, regular patching, firewalls, encryption, and continuous monitoring significantly improves the security of the system.

-) Phishing Awareness Campaign Report Analysis

Introduction:

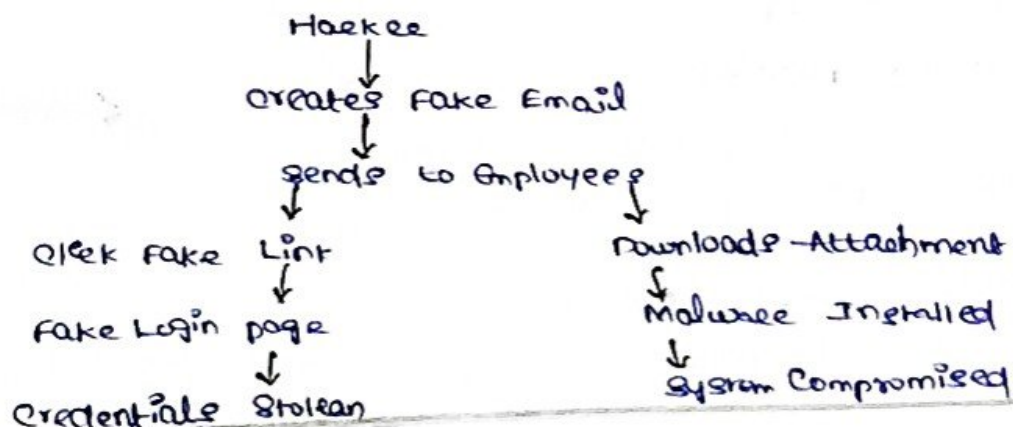
Phishing is a social engineering attack in which attackers trick users into revealing sensitive information such as usernames, passwords, and bank details. In this report, a phishing campaign was conducted for employees of ABC Technologies Pvt. Ltd.

Report findings

- 40% employees clicked the phishing link
- 20% entered their login credentials
- several employees downloaded a malicious attachment

These results indicate a low level of cybersecurity awareness among employees.

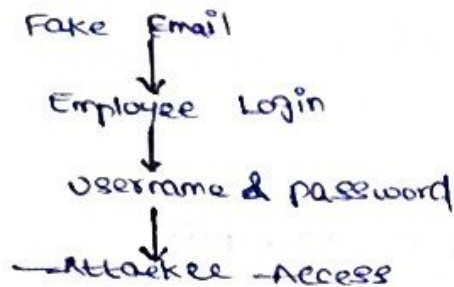
Phishing - Attack process diagram



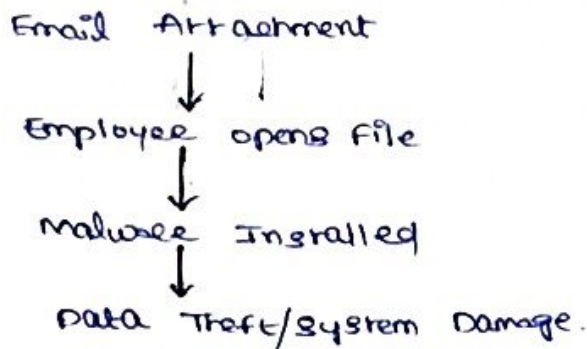
Security Weakness Identified

- 1) Employees clicked phishing links
- 2) credentials entered
- 3) Malicious attachment downloaded
- 4) NO email verification.

Credential Theft



Malware Attack



Security Recommendations

- 1) User Awareness Training
- 2) Email Security Controls
- 3) Multi-Factor Authentication
- 4) Strong password policy
- 5) Endpoint Security

Organizational Policies

- Regular cybersecurity awareness training.
- Mandatory phishing simulations.
- Email usage policy
- Incident reporting procedure
- Backup and recovery policy

Conclusion

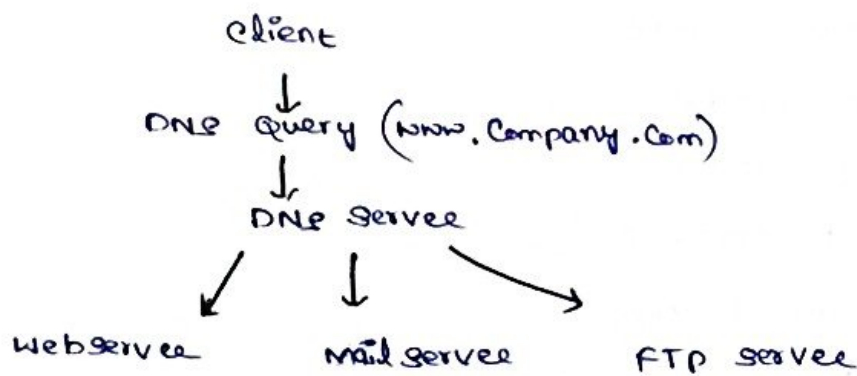
The Gophish Campaign showed that many employees were vulnerable to phishing attacks. Employee awareness, strong email security, MFA, and regular training can significantly reduce the risk of social engineering attacks and improve the organization's cybersecurity posture.

3) DNS Assessment Report Analysis

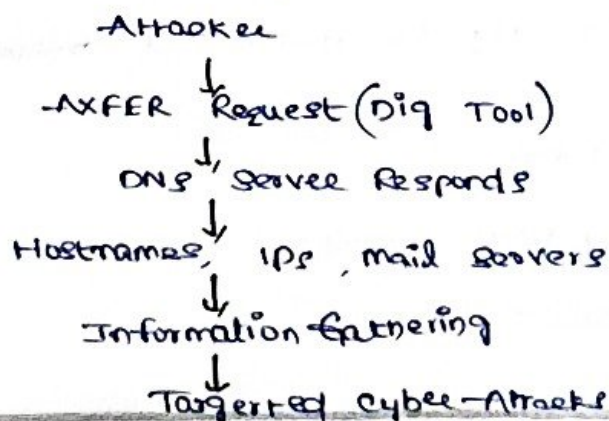
Introduction:

DNS (Domain Name System) translates domain names into IP addresses. A DNS zone transfer (AXFR) is used to copy DNS records b/w primary and secondary DNS servers. If AXFR is enabled for unauthorized users, attackers can obtain sensitive information about the organization's network.

DNS Architecture Diagram



DNS Zone Transfer - Attack



Security Implications

If AXFR is successful, attackers can discover

- Internal hostnames
- Mail Servers
- DNS records
- Web Servers
- Network Structure

This information can be used for:

- Footprinting
- phishing attacks
- Network mapping
- Targeted exploitation
- Lateral movement

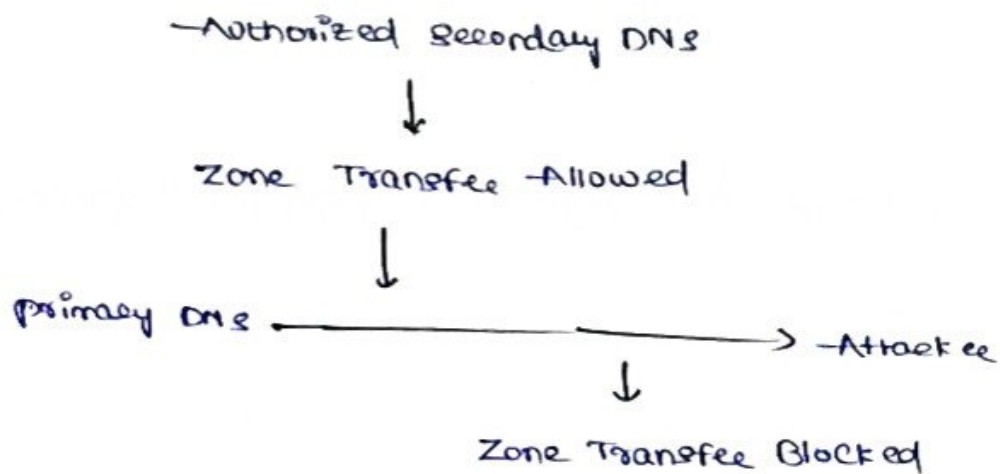
How Attackers Exploit the Information

- 1) Identify internal servers
- 2) Find vulnerable systems
- 3) Launch phishing attacks
- 4) Attack exposed services
- 5) Gain unauthorized access

Best practices to secure DNS

- Disable public AXFR
- Allow zone transfers only to authorized secondary DNS servers
- Configure firewall rules
- Remove unnecessary DNS records
- Restrict administrative access

Secure DNS Diagram



Conclusion:

→ A successful DNS zone transfer exposes valuable information about an organization's infrastructure increasing the risk of targeted cyberattacks. Restricting AXFR to authorized DNS servers implementing DNSSEC using firewalls and continuously monitoring DNS activity are essential best practices to secure DNS infrastructure and protect organizational assets.